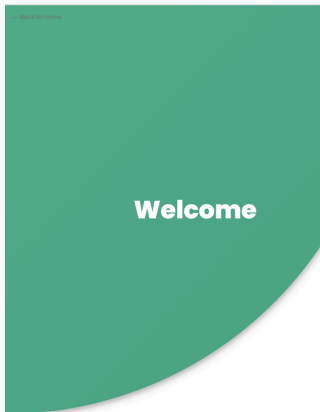




AI CHALLENGE IN SERVICE OF ISLAMIC CONTENT 2026

e-Tasmi AI Learning Loop

Pre-Challenge Baseline Documentation

Track 03 — Interactive Experiences & Learning Journeys for Islam

PARTICIPANT

Elhessan Abdelrahman Adam

STATUS

Pre-Challenge Source Baseline

Source snapshot: commit a358719

Tag: baseline-pre-ai-challenge-2026

This report records the system that already existed. It does not record challenge-period implementation.

Repository: <https://github.com/Elhessan2000/e-tasmi-ai-learning-loop>

Document Control

This report documents the pre-challenge source state. The source baseline is identified by the Git commit and tag below.

Item	Record
Project	e-Tasmi AI Learning Loop
Challenge	AI Challenge in Service of Islamic Content 2026
Track	Track 03 — Interactive Experiences & Learning Journeys for Islam
Participant	Elhessan Abdelrahman Adam
Repository	https://github.com/Elhessan2000/e-tasmi-ai-learning-loop
Branch	master
Baseline commit	a358719
Full commit	a35871918907d7a403c9469537c25f940009b340
Baseline tag	baseline-pre-ai-challenge-2026

The tag points at commit a358719. Later local inspection addresses are not part of that commit. They are recorded in Section 13 so the source snapshot is not confused with the machine setup used to view it.

This report documents the starting state. It is not a claim that challenge work has been accepted, completed, or scored.

1. Baseline Scope and Boundary

e-Tasmi already existed as a web platform for Qur'an recitation learning. This baseline records that system before the official challenge development period, so later work can be compared with a known starting point.

Pre-challenge baseline means the source in commit a358719 and the behaviour confirmed by the existing system audit, the application code, and the screenshots in this report. The project is being extended. It is not presented as a system created for the challenge.

Challenge-period implementation is outside this report. It will be documented separately, and only after that work is actually done.

Boundary	What it covers
PRE-CHALLENGE	Existing system and capabilities already present in the source snapshot.
CHALLENGE PERIOD	New work performed during the official challenge development period. Not described here as present.

The following were not part of the baseline and are not described as implemented: persistent structured AI findings; per-finding Accept, Edit, or Reject; an instructor verification gate for individual findings; Verified Learning Focus; a student learning-focus display; Practice Again; resubmission linkage; and a trusted Qur'an Library verse key used as the recitation comparison reference.

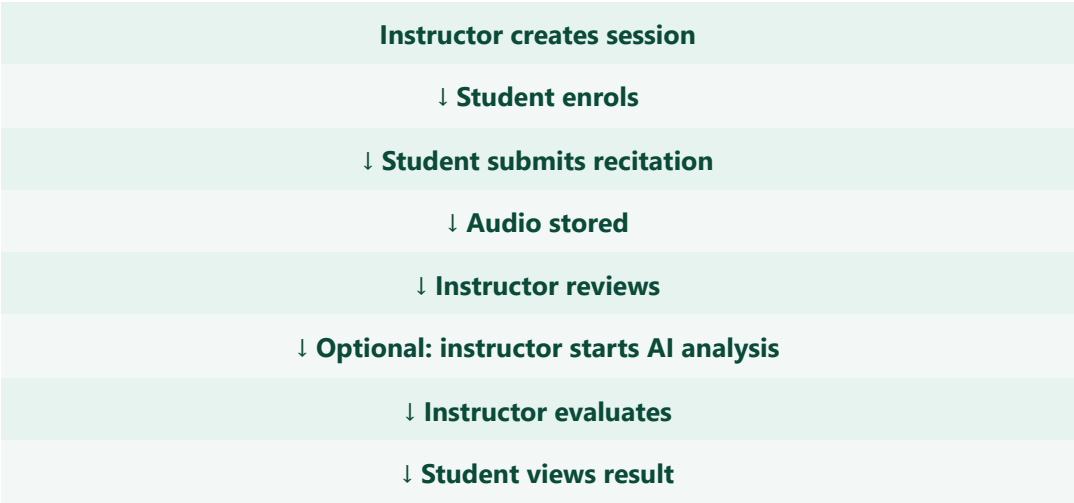
2. Existing System Overview

At baseline, e-Tasmi was a server-rendered JSP and Java Servlet application. A student enrolls in a tasmi session, submits recitation audio, and later views the instructor's score and feedback. An instructor manages sessions, plays submitted audio, may run an on-screen analysis, and saves the official evaluation. An administrator manages users and approves instructors.

Access is session-based. The audit records no separate course entity. The live path uses three roles only: Student, Instructor, and Admin.

Area	Already present
Learning unit	Tasmi session, including a free-text Qur'an portion
Enrolment	Student linked to a session
Recitation	Uploaded or recorded audio stored for that enrolment
Evaluation	One official score and feedback per recitation
Progress	Student progress updated from the saved evaluation
Qur'an Library	Reading, reference, and listening, separate from recitation comparison

The confirmed recitation path is shown below. AI analysis sits beside instructor review. It is optional, and only the instructor starts it.



3. Existing Student Experience

A student signs in, uses a dashboard, chooses a session, and submits audio by recording or by file upload. After the instructor saves an evaluation, the student opens the result, plays the submission, and reads the score and feedback. A submission can also remain waiting for review.

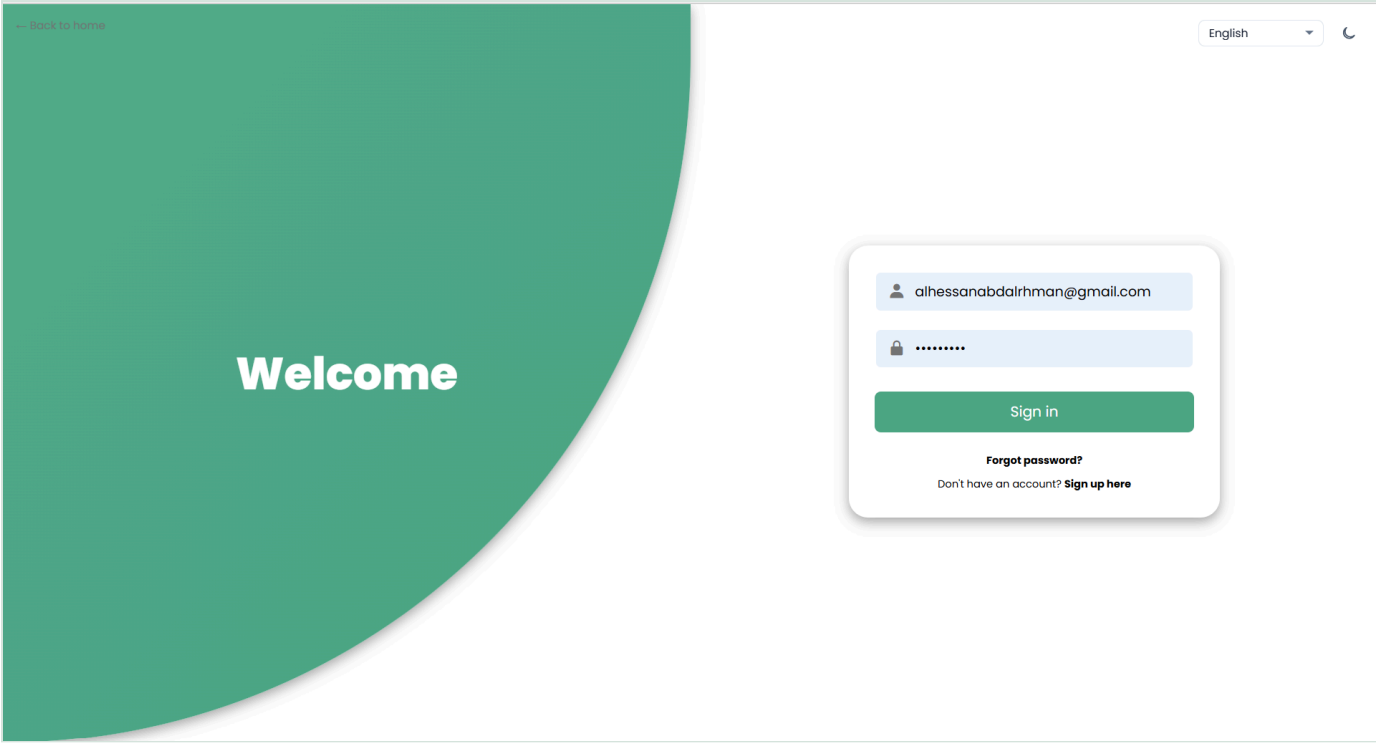


Figure 1. Existing login interface. The screen shows a welcome panel, email and password fields, Sign in, Forgot password, and Sign up.

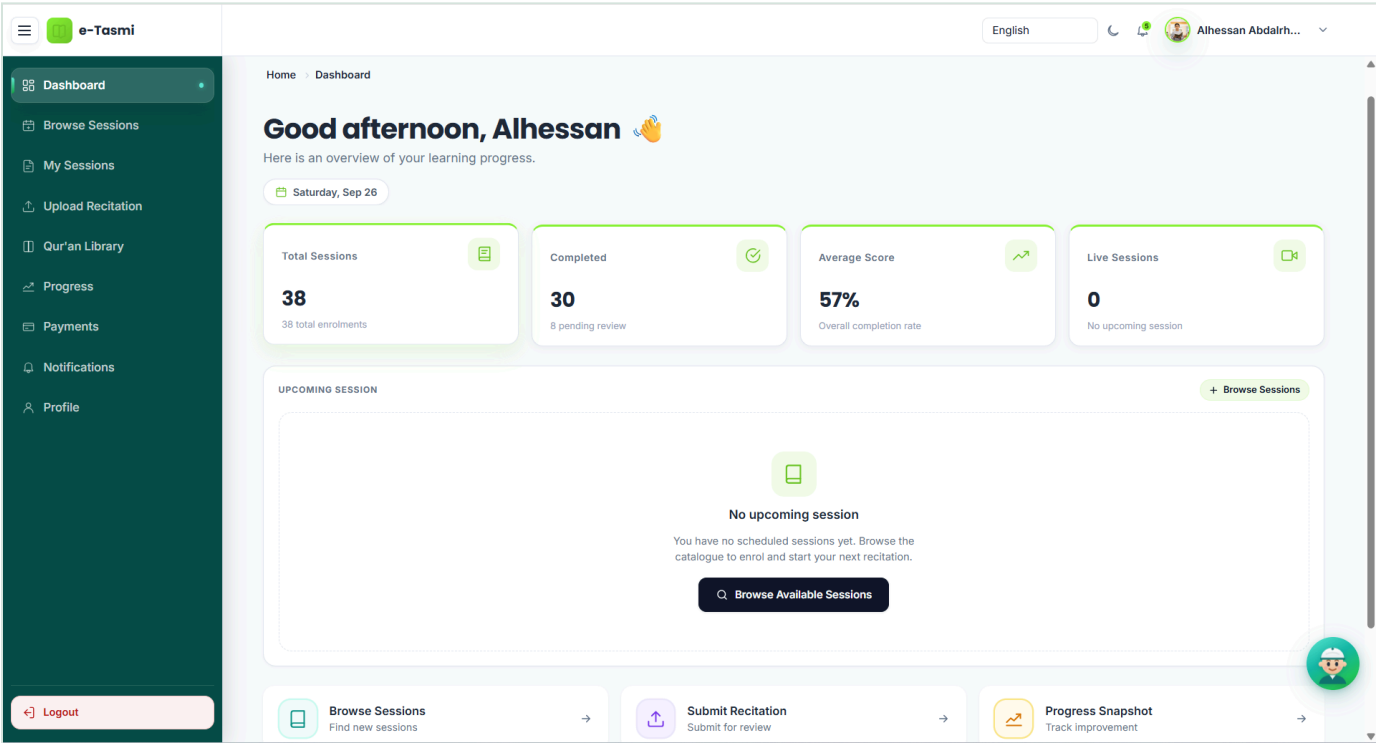


Figure 2. Existing student dashboard at the pre-challenge baseline. Visible items include session totals, completed sessions, an average score, and links to browse sessions, submit a recitation, and open progress.

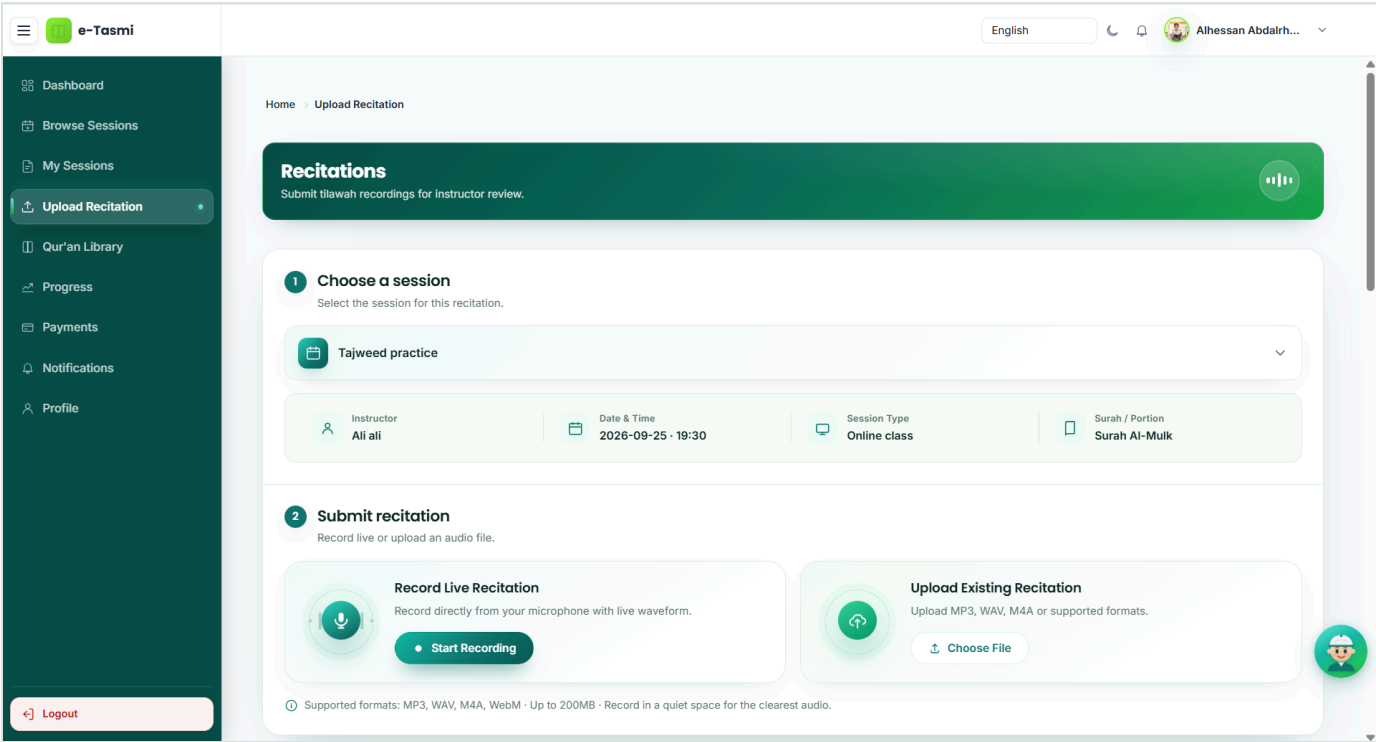


Figure 3. Existing student recitation submission interface. The screen shows session selection, the assigned surah or portion, Record Live Recitation, and Upload Existing Recitation.

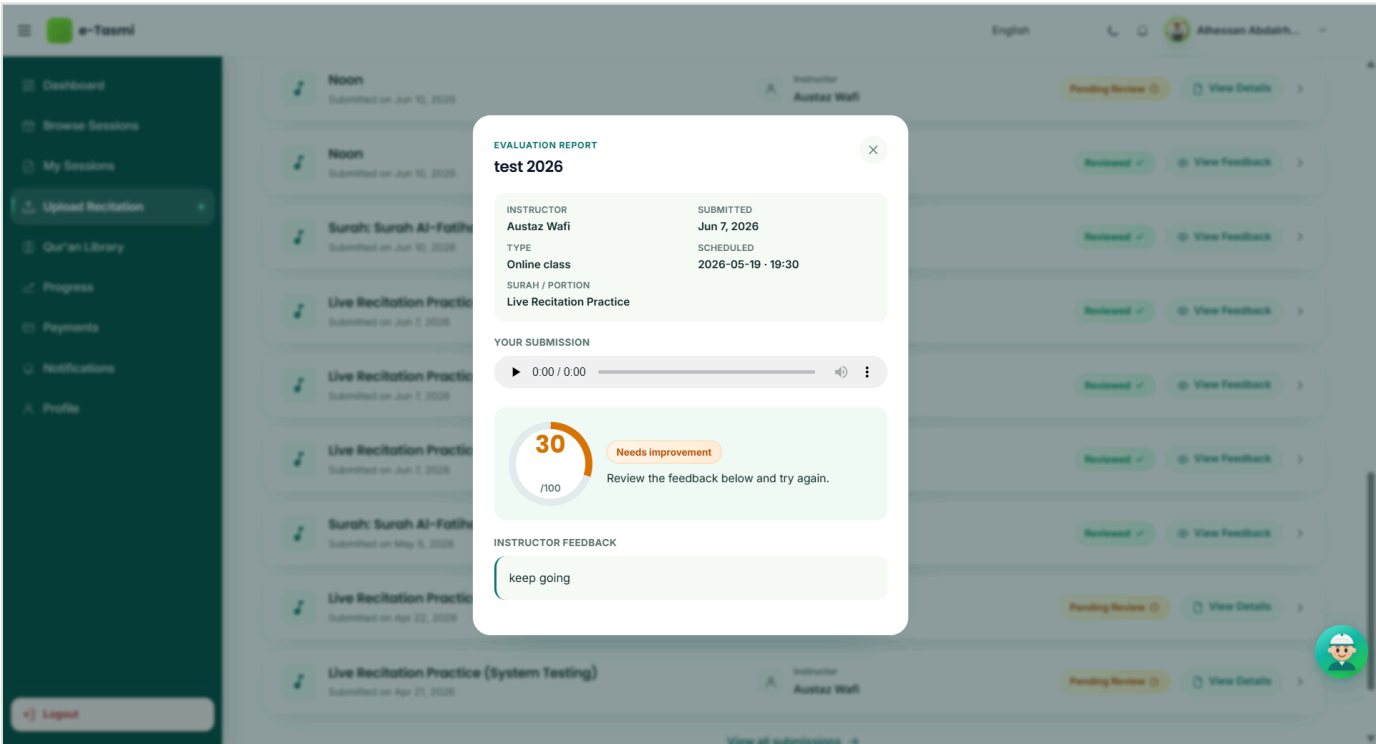


Figure 4. Existing student result and feedback interface. The open report shows playback, a score out of 100, a status, and an instructor feedback note.

Payment, notifications, and live join exist beside this path. They are not required for every recitation. The message on the result screen is the stored evaluation view. It is not the planned Practice Again learning loop.

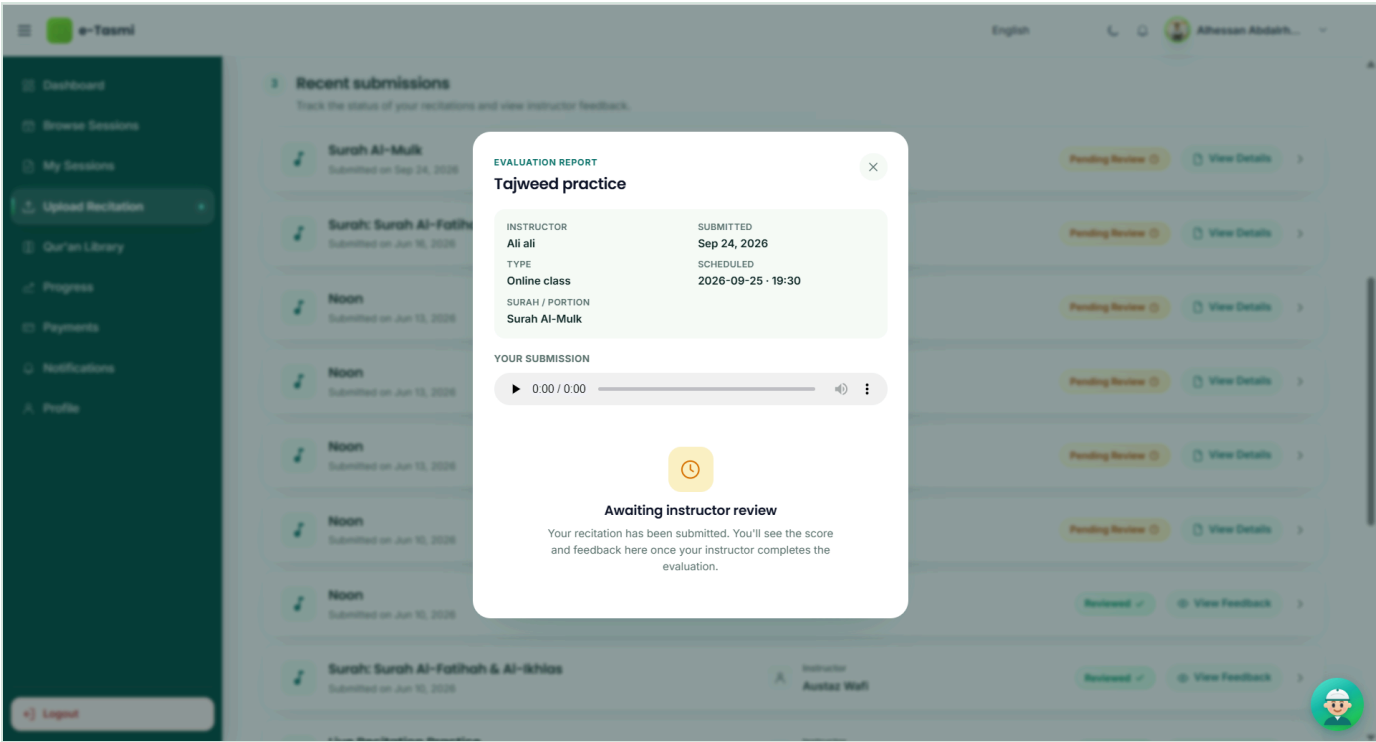


Figure 5. Existing student-side recitation workflow evidence. This report is waiting for instructor review and shows the session, portion, and submission playback.

4. Existing Instructor Experience

An approved instructor sees sessions, submissions, and a teaching calendar. On the evaluations page the instructor opens a session roster, plays submitted audio, and can start the existing analysis. The on-screen result is advisory. The instructor types the official score and feedback and saves them. That saved row is what the student sees.

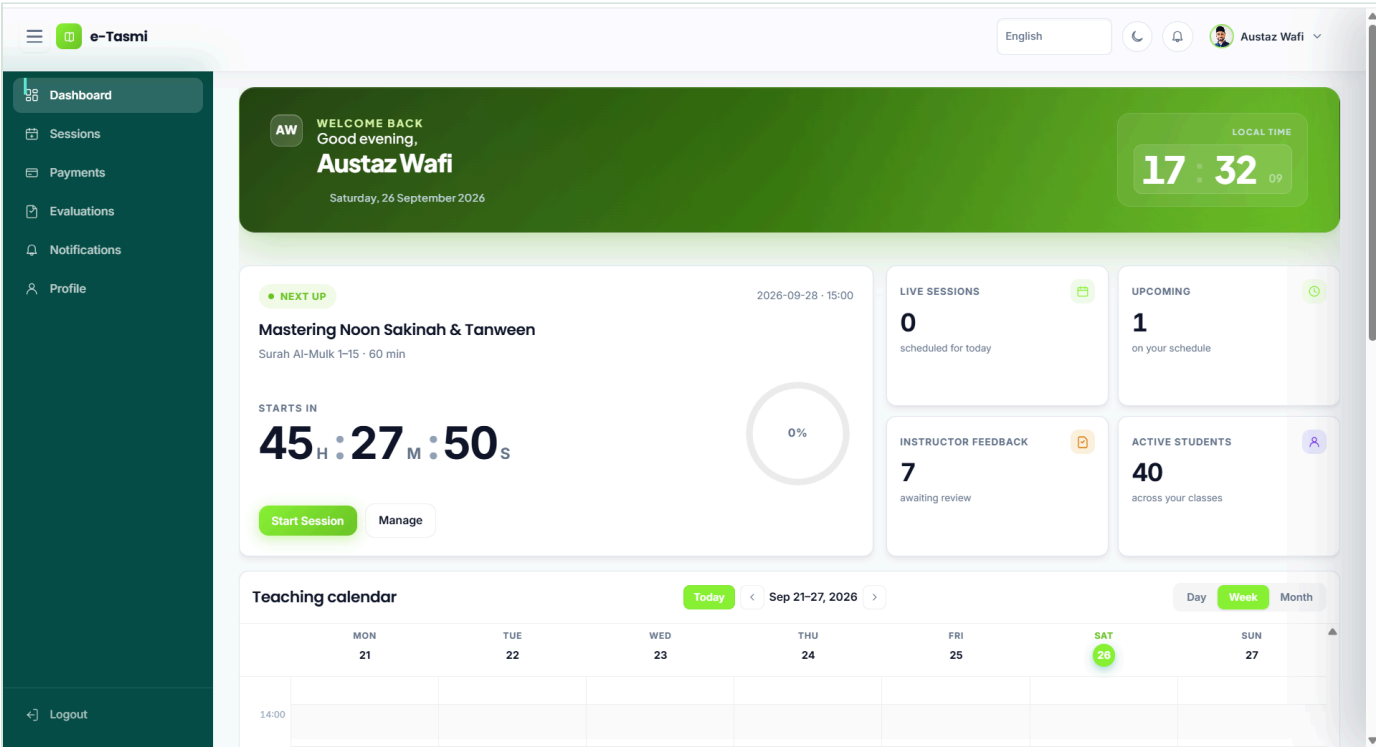


Figure 6. Existing instructor dashboard. Visible items include the next session, live and upcoming counts, feedback awaiting review, active students, and a teaching calendar.

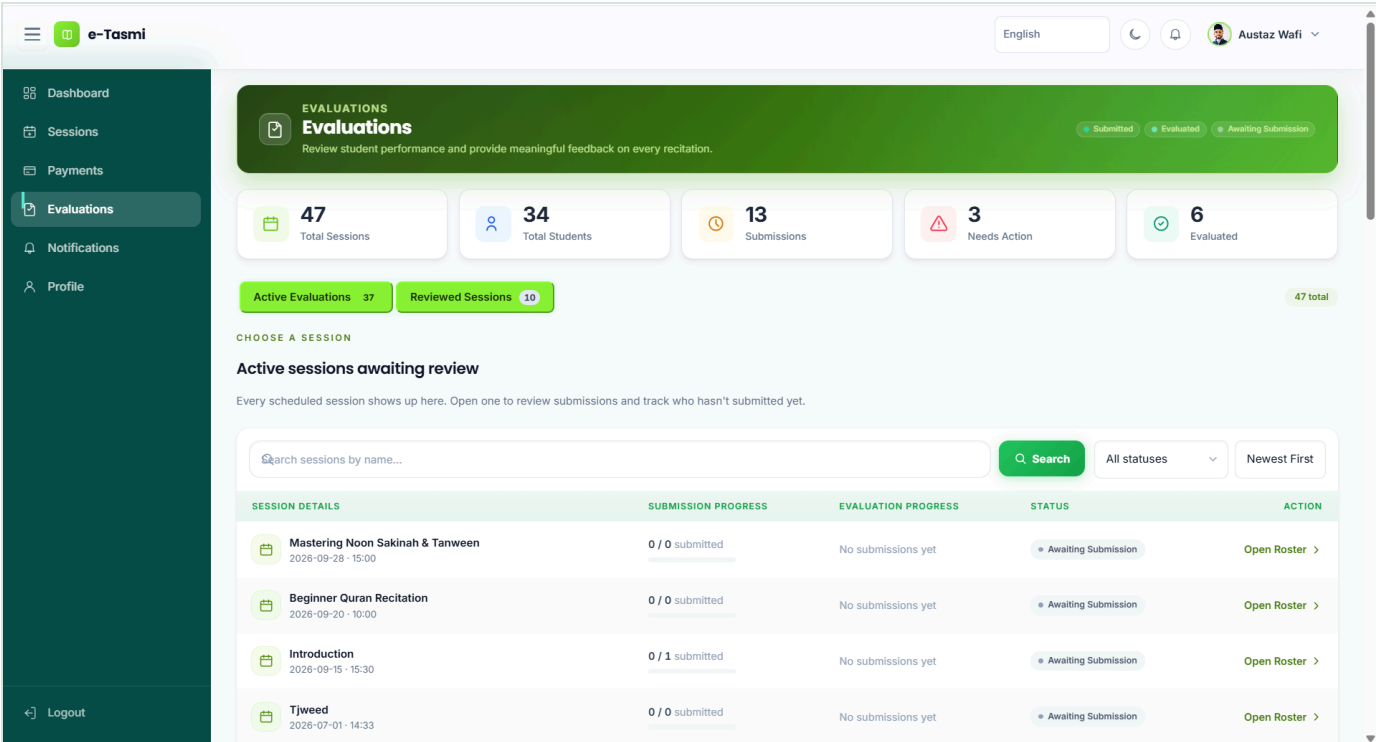


Figure 7. Existing instructor evaluation interface. The page lists sessions, submission progress, and evaluation status, with Active and Reviewed session groups.

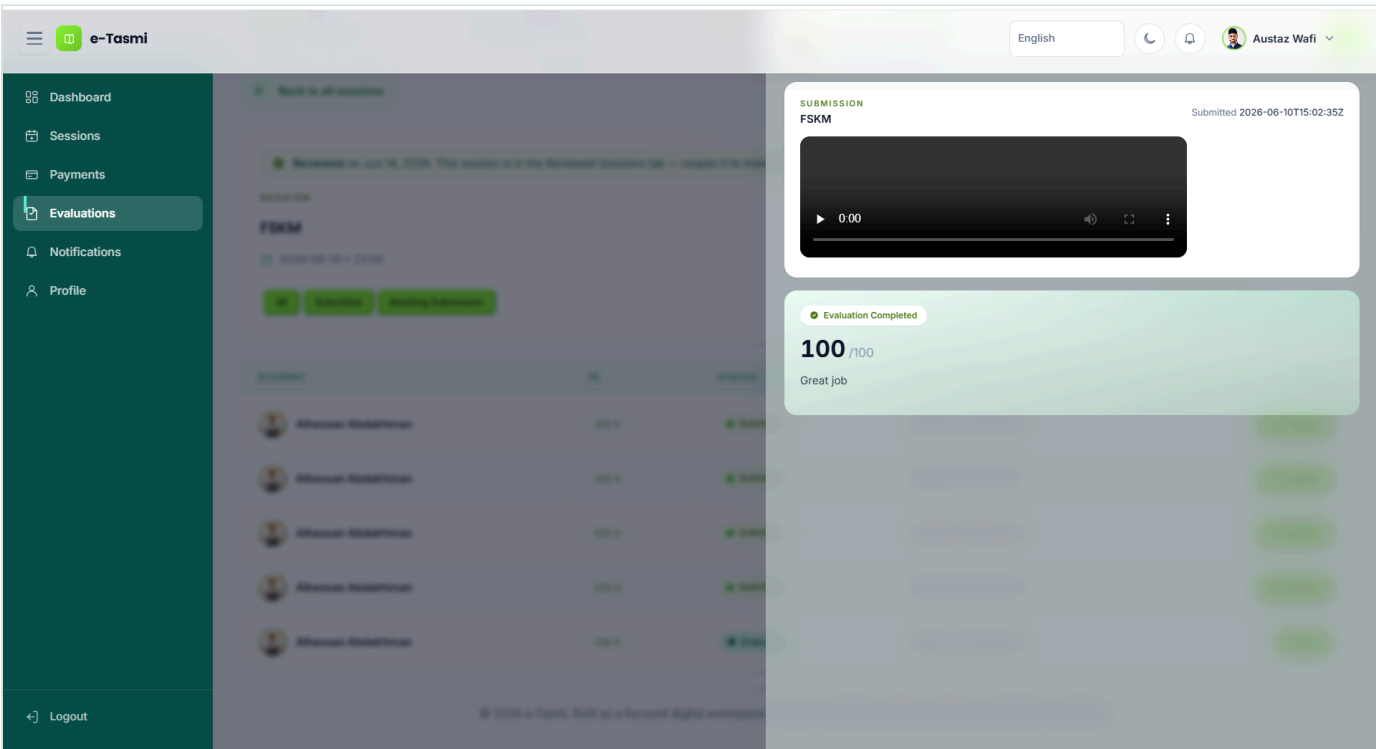
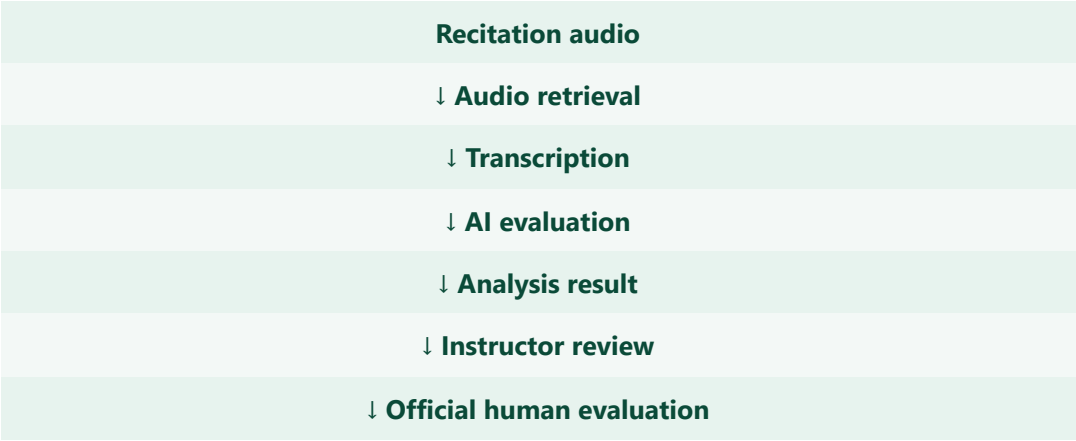


Figure 8. Existing instructor-side completed recitation workflow. The panel shows the submitted audio and a completed evaluation with a score out of 100 and feedback.

The score on the analysis report is a suggestion. The score stored for the student is the score the instructor enters and saves.

5. Existing AI-Assisted Recitation Analysis

Instructor-triggered analysis already existed. The student cannot start it. The service reads the stored audio, requests a transcription, then requests an evaluation against the session text. The result is shown on the evaluations page. It is not written as a durable structured finding. Saving the official evaluation clears that on-screen state, and a later page load does not restore it as a finding record.



A successful result can include a transcript, correct words, missing words, incorrect words, extra words, pronunciation notes, passage flags, a suggested score from 0 to 100, a summary, and feedback text. Pronunciation notes are limited to the transcript. The audit records no acoustic Tajweed engine. If the evaluator call fails, the same service can show a local word comparison instead.

The screenshot displays the 'e-Tasmi' application interface. On the left is a sidebar menu with options: Dashboard, Sessions, Payments, Evaluations (selected), Notifications, and Profile. The main area shows an 'Evaluations' section with a green header. Below the header, there are statistics: 47 Total Sessions, 34 Total Recitations, and 13 Evaluations. A 'Recitation Analysis Report' is shown for a student named 'FSKM', submitted on 2020-06-13T00:12:20Z. The report includes a video player showing a recitation, a score of 0/100 with a 'Needs Improvement' label, and a 'Passage mismatch' notice stating the student recited from Surah Al-Baqarah instead of the assigned lesson. It also shows word counts (Correct: 0, Missing: 1, Incorrect: 0, Extra / Repeated: 78), a transcript of the student's recitation, the expected text 'Noon', and a list of detected mistakes. The report concludes with a 'Passage Match' section stating the student recited from Surah Al-Baqarah instead of the assigned lesson, and an 'AI Summary' section summarizing the passage mismatch.

Figure 9. Existing instructor AI-assisted recitation analysis. The visible report includes playback, a suggested score, a passage-mismatch notice, word counts, a transcript, the assigned expected text, and a summary. The page tells the instructor to review the report before confirming a final score.

The comparison text is the session's *quran_portion* value: free text entered for that session, not a Qur'an Library verse key. On this screen the report labels that text as the assigned lesson and can flag a passage mismatch without turning the suggestion into the official score.

Java defaults, used only when the environment variables are absent, are gpt-4o-transcribe for transcription and gpt-4o for evaluation. Baseline Compose, when the host does not set the variables, passes whisper-1 for transcription and gpt-4o-mini on the classifier path. No runtime test of either pair is claimed here.

6. Existing Qur'an Library and Reference

The Qur'an Library already existed on the student path. Quran Foundation supplies chapters, verses, and related resources when configured. Reciter audio is served for listening. There is no local mushaf database in the application.

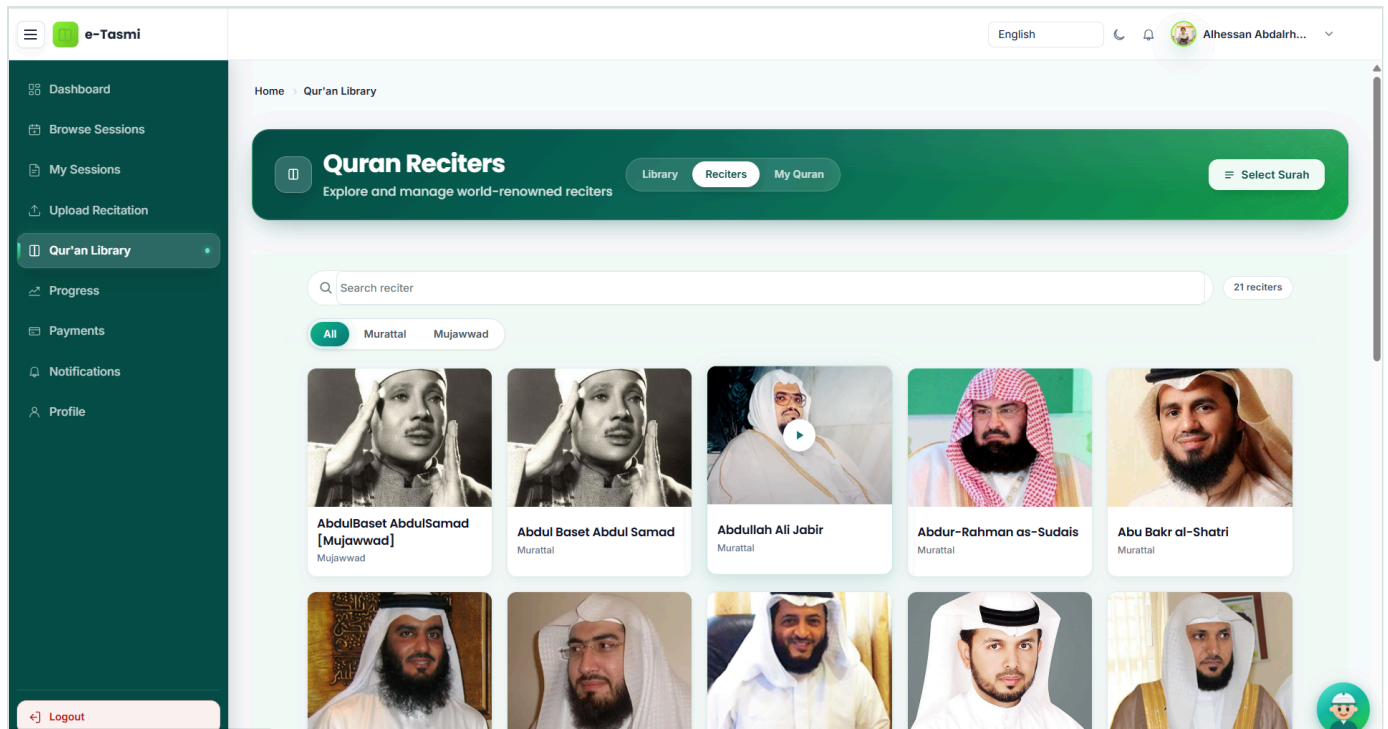


Figure 10. Existing Qur'an Library. The screen shows the reciters catalogue, search, and a play control.

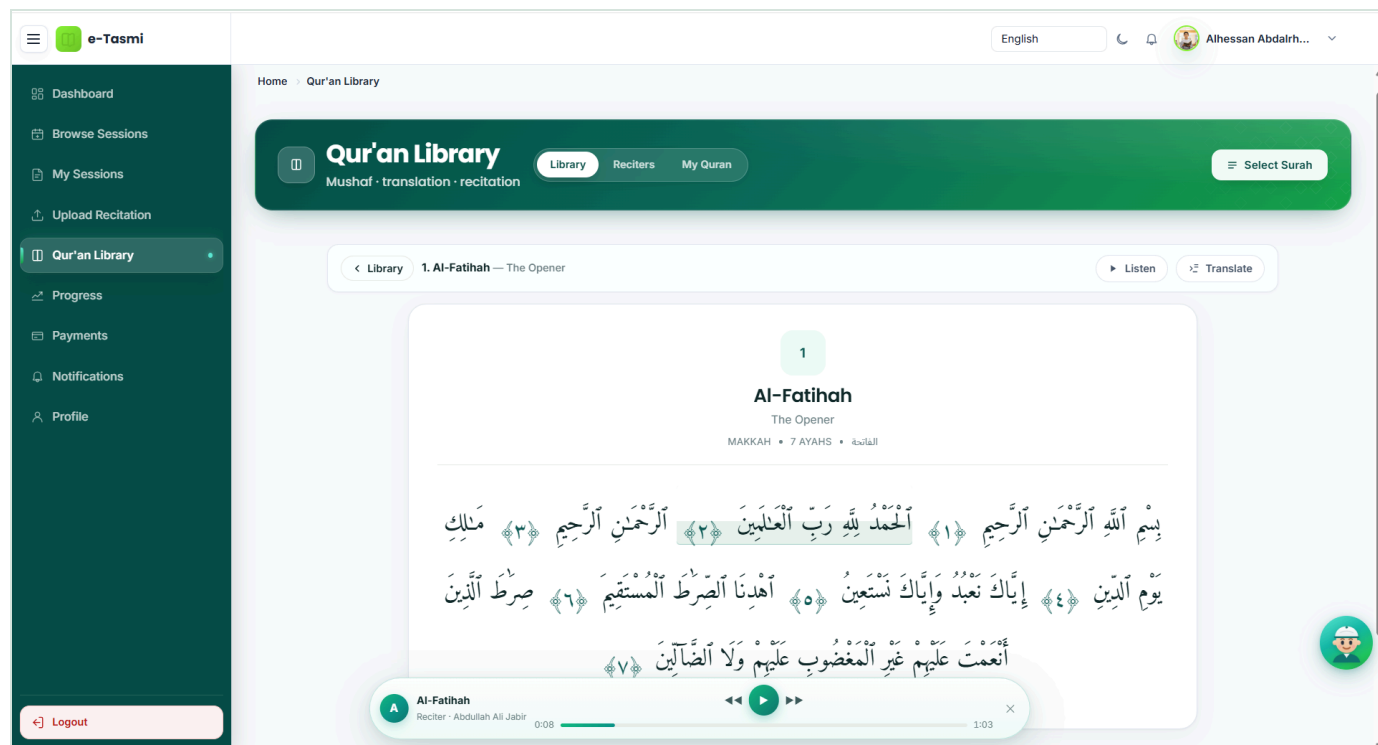
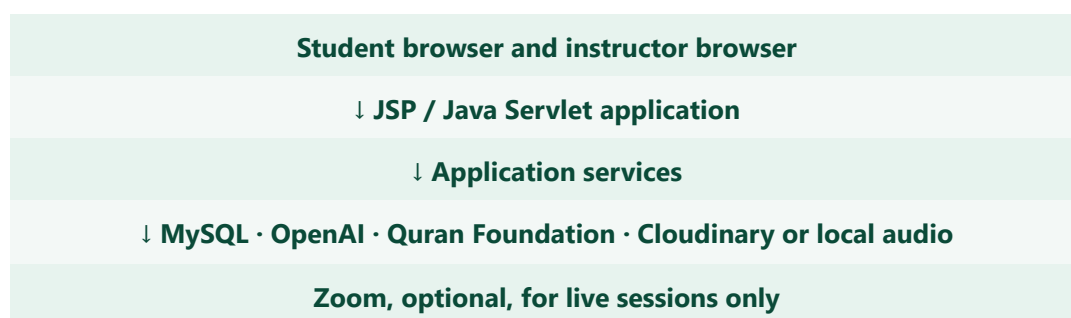


Figure 11. Existing Qur'an reference and library view. The screen shows Surah Al-Fatihah in Arabic, Listen and Translate controls, and a reciter player.

At baseline, the Qur'an Library reference path was separate from the recitation AI comparison path. The recitation comparison used the session's free-text quran_portion rather than a Qur'an Library verse key.

7. Existing System Architecture

The source is a servlet and JSP application with controller, service, and JDBC data-access code. This is a functional view. The audit does not define a separate formal framework. Not every external service is required for every feature.



MySQL stores accounts, sessions, enrolments, recitations, evaluations, and progress. OpenAI is used for the instructor analysis and, separately, for the student assistant, voice, and translation fallback when those features are configured. Quran Foundation supplies library content. Audio is read back from Cloudinary or from local /uploads for playback and for analysis.

The database name is etasmi. Inside Docker the application uses jdbc:mysql://db:3306/etasmi. Compose initialises a new database volume from e-Tasmi/e-Tasmi/setup/etasmi_schema.sql. The application can also apply its bundled schema script when the user table is missing.

8. Existing Data Model

The relationships below are foreign keys in the baseline schema. Challenge-specific tables are omitted. No durable AI-findings table existed.

From	Relationship	To
user	one profile	student or instructor
instructor	owns	tasmi_session
student + session	unique pair	enrollment
enrollment	holds	recitation
recitation	one official row	evaluation
instructor	records	evaluation
student	one row	progress

evaluation.recitation_id is unique, so one official evaluation is stored per recitation. *progress* uses the student as its key. The audit records that progress is updated when an evaluation is saved. *quran_portion* is a free-text column on *tasmi_session*. It is not a foreign key to a verse.

9. Technical Environment

Component	Version or technology	Role
Java	17	Application runtime and build
Apache Tomcat	9	Web and application server
MySQL	8.4	Relational database
JSP	Existing	Server-rendered interface
Java Servlets	Existing	Request handling
Docker	Existing	Containerization
Docker Compose	Existing	Local multi-container environment
Caddy	Image caddy:2-alpine	Optional reverse proxy
phpMyAdmin	Image phpmyadmin:5-apache	Local database inspection

Service	Baseline role
OpenAI	Instructor transcription and evaluation; separate student assistant features
Quran Foundation	Qur'an Library content when configured
Cloudinary	One audio and upload store; local /uploads is the other path
Zoom	Optional live sessions; a manual link or non-live mode can be used instead

Environment configuration can override the Java model defaults. Baseline Compose sets `OPENAI_RECITATION_MODEL` to `whisper-1` and `OPENAI_CLASSIFIER_MODEL` to `gpt-4o-mini` when the host does not supply other values. It does not set `OPENAI_EVALUATOR_MODEL`, so that classifier value is

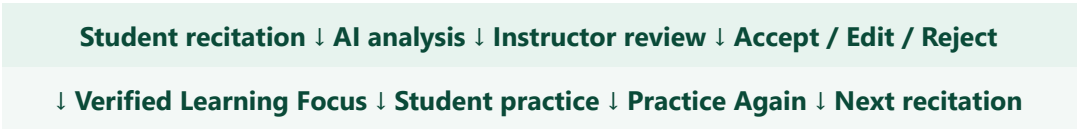
the evaluation path. gpt-4o-transcribe and gpt-4o remain the defaults in source when those variables are absent. This report does not claim a runtime measurement of either pair.

10. Existing System vs Planned Challenge Contribution

The challenge column is a plan. It is not a record of finished work.

Capability	Baseline	Planned contribution
Student recitation submission	Existing	Extend and reuse
Instructor evaluation	Existing	Extend and reuse
AI-assisted analysis	Existing; instructor-triggered	Extend
Structured AI findings	Partially existing on screen only	Persist as structured records
Trusted Qur'an reference	Existing in the library; not the comparison path	Connect to recitation analysis
Instructor Accept	Not present	Planned
Instructor Edit	Not present	Planned
Instructor Reject	Not present per finding	Planned
Instructor verification	Not present per finding	Planned
Verified Learning Focus	Not present	Planned
Student learning-focus display	Not present	Planned
Practice Again	Not present	Planned
Resubmission linkage	Not present	Planned
Continuous learning loop	Not present	Planned

PLANNED CHALLENGE-PERIOD FLOW



At baseline the path stops at instructor review of an on-screen analysis, an instructor-entered score and feedback, and the student viewing that stored result.

11. Third-Party Components, Rights and Permissions

Roles below are confirmed. Licence names are not stated from memory. Where terms have not been collected, the status remains To be verified.

Component	Baseline use	Licence or terms
OpenAI	Recitation transcription and evaluation; student assistant features	To be verified
Quran Foundation	Library chapters, verses, and related resources	To be verified
Cloudinary	Audio and other uploads, with local storage as the alternative	To be verified
Zoom	Optional live sessions	To be verified
MySQL	Database server	To be verified
Apache Tomcat	Application server	To be verified
Docker and Compose	Local containers	To be verified
phpMyAdmin	Local database inspection	To be verified

The e-Tasmi application is pre-existing project work. The participant must have the rights needed to submit and publish the repository. That requirement is not a record that those rights have already been confirmed. University, employer, and other ownership questions are To be verified. No ownership agreement is stated here. No root LICENSE file was found. Images, audio, fonts, icons, and front-end libraries still need a source and licence check before final submission.

12. Data Privacy and Public Repository Handling

The public repository must not contain API keys, passwords, access tokens, private credentials, real beneficiary records, or other personal records. This is a preparation rule, not a certification.

NO SECRETS
NO PRODUCTION DATA
NO REAL BENEFICIARY RECORDS

A local .env file is ignored by Git. .env.example is the tracked template and uses placeholders. This repository's current history contains one commit, a358719. .env is not in that history and is not tracked. That check applies only to this repository's current history.

The schema script is included for reproducibility. Production dumps and real user recordings are not part of this documentation. uploads/ and mysql-data/ are ignored. Demo material should be synthetic or anonymized. Screenshots still require a manual privacy review for names, email addresses, phone numbers, payment details, passwords, and tokens. This report does not claim that review is complete. External-service retention terms must be checked against the providers' current terms. They are not stated here.

13. Git Baseline and Source Integrity

Item	Value
Branch	master
Baseline commit	a358719
Full hash	a35871918907d7a403c9469537c25f940009b340
Commit message	chore: establish pre-challenge baseline
Tag	baseline-pre-ai-challenge-2026

The tag points at that commit. The baseline history has not been rewritten, and the baseline commit has not been amended. Later challenge work will be represented by subsequent commits, not by moving the tag.

The source snapshot itself was clean when it was created. The working tree used to prepare this report also contains later uncommitted isolation edits and untracked documentation. Those files are not part of commit a358719.

Later local inspection environment — not part of the baseline source commit

Application: http://localhost:8084

phpMyAdmin: http://localhost:8083

MySQL host port: 3317

These host ports were introduced after a358719 so this copy could run beside the original installation. The snapshot itself publishes MySQL on 3306, the application on 8080, Caddy on 8443, and phpMyAdmin on 8081.

14. Baseline Evidence Index

The eleven screenshots below are the interface evidence embedded in this report. Filenames match the files in documentation/baseline/screenshots/.

File	What it demonstrates
login.png	Existing login interface
student-dashboard.png	Existing student dashboard
recitation-submission.png	Existing student recitation submission
student-recitation-result.png	Existing student result and feedback
instructor-dashboard.png	Existing instructor dashboard
instructor-evaluation.png	Existing instructor evaluation interface
existing-ai-analysis.png	Existing instructor AI-assisted recitation analysis
quran-library.png	Existing Qur'an Library
quran-reference.png	Existing Qur'an reference/library view
existing-recitation-workflow-student.png	Existing student-side recitation workflow
existing-recitation-workflow-instructor-completed.png	Existing instructor-side completed recitation workflow

Supporting baseline documents, kept with this report: baseline declaration; existing system; existing AI capabilities; existing system versus planned contribution; architecture; technical environment; Git baseline; third-party rights; data privacy; screenshot index; evidence index. The evidence folder is an index for material still to be collected. It does not hold a separate evidence file beyond that index.

15. Baseline Conclusion

Commit a358719 captures a working pre-challenge e-Tasmi platform. It already supports recitation submission, instructor evaluation, instructor-side AI assistance, a Qur'an Library and reference view, and the supporting Java, Tomcat, MySQL, and Docker environment.

The AI assistance is instructor-triggered, shown on screen, and compared with the session's free-text quran_portion. The library path is separate. Official scores and feedback are entered by the instructor. The planned learning loop, including per-finding review and Practice Again, is not part of this baseline.

The challenge contribution is intentionally separated from this baseline. It will be documented through later implementation commits and evidence. This report makes no competitive claim.

Baseline commit: a358719

Baseline tag: baseline-pre-ai-challenge-2026

Appendix

Primary sources for this report are the baseline Markdown set in documentation/baseline/, the schema file e-Tasmi/e-Tasmi/setup/etasmi_schema.sql, RecitationAiAnalysisService, InstructorEvaluationServlet, and the existing system audit in the repository. No secret values are included.